Before class: twenty building blocks

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. The answer key is at the back.

1 · Read first (5 min)

An acid and a base in one

An **amino acid** has an amine (–NH_2, a base) and a carboxylic acid (–COOH) on the same carbon, plus a side chain. In water the acid gives its proton to the amine of the same molecule, so amino acids carry both a + and a − charge.

The amide bond, again

In Chapter 4 an acid and an amine joined, losing water, to make an **amide**. Amino acids do the same, over and over: the chain of amide (peptide) bonds is a **protein**. Insulin has 51 amino acids; many proteins have hundreds.

Shape is everything

A protein chain folds into a precise 3D shape. Heat, alcohol or heavy metals can unfold it (**denaturation**), as when an egg white turns solid. **Enzymes** are proteins whose shape lets them speed up one reaction; many drugs work by blocking an enzyme.

2 · Readiness check

Ten quick questions: recap and a preview.

Q1 Recap

A carboxylic acid + an amine, heated, gives:

A. an amide + water
B. an ester + water
C. an ether
D. a ketone

Q2 Recap

Which group is basic?

A. $-NH_2$
B. –COOH
C. –OH
D. –CHO

Q3 Recap

Two cysteine –SH groups can be oxidised to form a:

A. disulfide bridge
B. peptide bond
C. ester
D. glycosidic bond

Q4 CH7-LO1

How many different amino acids are commonly found in proteins?

A. 20
B. 4
C. 64
D. 100

Q5 CH7-LO1

All amino acids in human proteins are the L form.

True / False

Q6 CH7-LO2

A molecule with both a + and a − charge but no overall charge is a:

A. zwitterion
B. radical
C. racemate
D. polymer

Q7 CH7-LO3

Proteins are polymers of:

A. amino acids
B. glucose
C. fatty acids
D. nucleotides

Q8 CH7-LO4

Frying an egg turns the white solid because the protein is:

A. denatured
B. hydrolysed to amino acids
C. oxidised to sugar
D. saponified

Q9 CH7-LO5

Enzymes are best described as:

A. biological catalysts, mostly proteins
B. hormones made of fat
C. sugars
D. vitamins

Q10 CH7-LO5

Many medicines work by blocking an enzyme. Name one you have heard of, or suggest why blocking an enzyme could treat a disease.

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PHD125 Pharmaceutical Organic Chemistry · Amino Acids, Proteins and Enzymes

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Answer key

2 · Readiness check

Q1 Recap

A carboxylic acid + an amine, heated, gives:

A. an amide + water

B. an ester + water

C. an ether

D. a ketone

Chapter 4.

Q2 Recap

Which group is basic?

A. $-NH_2$

B. –COOH

C. –OH

D. –CHO

The N lone pair accepts H^+ (Chapter 4).

Q3 Recap

Two cysteine –SH groups can be oxidised to form a:

A. disulfide bridge

B. peptide bond

C. ester

D. glycosidic bond

Chapter 3.

Q4 CH7-LO1

How many different amino acids are commonly found in proteins?

A. 20

B. 4

C. 64

D. 100

The same 20 in all living things.

Q5 CH7-LO1

All amino acids in human proteins are the L form.

True / False → **True**

Glycine is not chiral; all the others are L.

Q6 CH7-LO2

A molecule with both a + and a − charge but no overall charge is a:

A. zwitterion

B. radical

C. racemate

D. polymer

Amino acids in water.

Q7 CH7-LO3

Proteins are polymers of:

A. amino acids

B. glucose

C. fatty acids

D. nucleotides

Joined by peptide bonds.

Q8 CH7-LO4

Frying an egg turns the white solid because the protein is:

A. denatured

B. hydrolysed to amino acids

C. oxidised to sugar

D. saponified

Its 3D shape is destroyed.

Q9 CH7-LO5

Enzymes are best described as:

A. biological catalysts, mostly proteins

B. hormones made of fat

C. sugars

D. vitamins

They speed reactions without being used up.

Q10 CH7-LO5

Many medicines work by blocking an enzyme. Name one you have heard of, or suggest why blocking an enzyme could treat a disease.

Examples: statins block HMG-CoA reductase to lower cholesterol; aspirin blocks COX to reduce pain and inflammation; penicillin blocks a bacterial enzyme that builds the cell wall. Blocking the enzyme stops the harmful product (or the microbe's essential product) being made.

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